Problem 2.6

Suppose that the accumulation function for an account is

$$a(t) = (1 + 0.5it).$$

You invest \$500 in this account today. Find i if the account's value 12 years from now is \$1,250.

Solution

The accumulated value after 12 years is given by

$$1250 = 500(1 + 0.5i \cdot 12).$$

Simplify the factor inside the parentheses:

$$1250 = 500(1 + 6i).$$

Divide both sides by 500:

$$\frac{1250}{500} = 1 + 6i.$$

So,

$$2.5 = 1 + 6i.$$

Then,

$$2.5 - 1 = 6i,$$

which gives

$$1.5 = 6i.$$

Therefore,

$$i = \frac{1.5}{6} = 0.25.$$

Hence,

$$\boxed{i = 25\%}.$$